Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	In-situ Product Removal for Higher Productivity in Bio-based Organic Acid Fermentation – Demonstrated by Succinic Acid			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes/ Bio-Equipment	Bio-process Technologies	Industrial Biotechnology	Bio-Chemical Materials

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Concept of Proposed Technology	The proposed technology is an integrated fermentation process with in-situ product removal (ISPR) for organic acids, exemplified by succinic acid. Microorganisms convert glucose or other sustainable substrates and CO_2 into succinic acid in a stirred bioreactor, while the product is selectively withdrawn from the broth (e.g. by membranes, adsorption, or extraction). This coupling of bioconversion and product removal keeps concentrations below inhibitory levels, stabilizes the culture, and increases space-time yields while reducing downstream burden. Key functions are microbial conversion of renewable substrates, in-situ removal and partial concentration of succinic acid, and mitigation of product inhibition and pH stress. The ISPR unit is designed to selectively capture the product while retaining carbon substrates and essential nutrients in the reactor to maintain productivity and avoid extra nutrient addition. In future implementations, automated control of product removal based on biological signals, such as characteristic pH changes during organic acid formation, is envisaged to trigger ISPR cycles at optimal times. Overall, the reactor operates as a combined production–separation unit that delivers an enriched succinic acid stream for conventional downstream purification, and the same concept can be adapted to other organic acids or bioproducts. Potential uses include bio-based chemicals and bioplastics (e.g. PBS, plasticizers, resins), where integrated ISPR can lower downstream costs, enable continuous or intensified operation, and handle inhibitory product levels. The lab-scale ISPR setup also serves as a flexible R&D platform at the interface of microbiology, separation science, and process engineering, and can be extended beyond succinic acid to a broader product portfolio.
Rationale for Joint R&D Project	Fraunhofer UMSICHT offers an excellent basis for this joint project: the institute has many years of experience in fermentation, including processes for succinic acid production, and operates a broad infrastructure ranging from lab-scale to pilot-scale reactors. This allows systematic development, testing and upscaling of fermentation processes under realistic conditions. However, despite this strong fermentation and scale-up competence, a crucial building block is missing: an efficient, tailor-made in-situ product removal (ISPR) system for succinic acid and related organic acids. This is precisely where the Korean partner's expertise becomes indispensable. The Korean team should contribute advanced know-how in separation technologies and develop dedicated ISPR solutions (e.g. membrane-, adsorption- or extraction-based modules) optimized for organic acid removal. Fraunhofer UMSICHT will integrate these modules into its succinic acid fermentations, evaluate their impact on productivity and process stability, and design control strategies that enable (semi-)automated product removal, for example triggered by biological signals such as characteristic pH changes. The collaboration is therefore highly synergistic and strategically important: by combining Korea's

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	expertise in separation technologies with Fraunhofer UMSICHT's proven strengths in fermentation and multi-scale process development, the partners aim to develop and demonstrate an integrated ISPR–fermentation system and explore options for (semi-)automated operation within the 3-year project period.
Key Words	1. In-situ product removal (ISPR) 2. Organic acid-selective separation 3. Succinic acid bioproduction 4. (Semi-)automated, biologically triggered product withdrawal 5. Process development

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	**Selective ISPR modules for succinic and related organic acids** The project will develop and optimize practically relevant in-situ product removal (ISPR) modules based on innovative separation principles. These modules will be tailored for the selective removal of succinic acid and related organic acids from fermentation broth, with minimal co-removal of carbon substrates and essential nutrients, and will be designed for sterile, CIP/SIP-compatible operation. **Integrated fermentation–ISPR process loop at lab scale** A lab-scale fermentation process will be established in which the bioreactor is coupled to the Korean ISPR modules via a side-stream loop. Operating strategies (continuous or cyclic product withdrawal) will be developed and validated that increase achievable titers and space-time yields and reduce the need for neutralizing agents, while maintaining culture stability. The outcome will be a robust, experimentally validated reference system for integrated organic acid production with in-situ product removal. **Robust, biology-based monitoring and (semi-)automated ISPR operation** Simple, robust monitoring and control concepts will be developed that use process signals such as pH and conductivity to trigger and adjust ISPR operation. The aim is to demonstrate (semi-)automated product removal in the integrated system, using rule-based strategies that can be readily transferred to other organic acid fermentations.			
R&D Target Specifications	At lab scale, the project aims to demonstrate an integrated succinic acid fermentation with in-situ product removal (ISPR) that clearly improves on the reference process at Fraunhofer UMSICHT. Over the course of the project, the achievable operating window is to be significantly expanded, i.e. distinctly higher titers and productivities are to be reached while maintaining or improving yield on substrate. The ISPR unit will be designed to selectively remove succinic acid from the broth, with only minor co-removal of fermentable sugars and essential nutrients, and to achieve a clearly measurable reduction in neutralizing agent demand and salt formation compared to the non-ISPR benchmark. The integrated system is expected to operate stably on a weekly basis in continuous or repeated-batch mode, and a substantial proportion of ISPR operation should be triggered by simple, biology-based rules (e.g. using pH or related process signals), thereby demonstrating the practical feasibility of (semi-)automated control.			

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R&D Plan	**Months 1–6 (Pre-study: separation focus with synthetic solutions)** Jointly review and prioritization of suitable separation concepts for the in situ removal of succinic acid and related organic acids (e.g. membrane-based, adsorption/ion exchange, extraction). The process and material requirements are defined, and the necessary development and optimization steps are discussed. The result is a shortlist of 2–3 promising innovative separation concepts and a joint specification that links biological requirements (titer, pH value, sterility, acceptable losses) with the objectives of the separation design. **Months 7–18 (Year 1 main phase: module development and reference fermentation)** Based on the pre-study, the Korean partner designs 1–2 concrete ISPR module types and characterizes them using synthetic solutions and clarified fermentation broths, focusing on selectivity, capacity, stability and cleanability. In parallel, Fraunhofer UMSICHT establishes a robust, reproducible lab-scale succinic acid reference fermentation (1–5 L stirred tanks) and quantifies key indicators (titer, productivity, yield, base consumption, inhibition thresholds). At the end of this phase, 1–2 ISPR module designs are selected and the biological reference process is well defined. **Months 19–30 (Year 2: integration and (semi-)automation)** Fraunhofer UMSICHT integrates the selected ISPR modules into a lab-scale fermentation via a side-stream loop with pumps, valves and basic sensors (e.g. pH, flow, pressure, conductivity). Systematic experiments compare operating strategies (continuous vs. cyclic product removal, varying side-stream flow rates and trigger points). The impact of ISPR on titers, productivities, yields, base consumption and culture stability is quantified. With strong contact to the Korean partner iterative optimization of the ISPR modules are performed. Simple, robust rule-based control strategies are implemented, where e.g. pH (and optionally conductivity or at-line acid measurements) are used to start/stop or adjust ISPR operation, demonstrating practical (semi-)automation. **Months 31–36 (Year 3: optimization, modelling, platform concept)** The integrated system is operated to test stability and reproducibility under optimized conditions. Simple dynamic mass-balance models for the bioreactor and ISPR loop are developed and calibrated to support understanding and optimization (e.g. influence of removal rates, trigger thresholds, module sizing). From experiments and modelling, we derive initial design and operating guidelines and formulate a platform concept that specifies for which organic acids and parameter ranges the approach is suitable and how it could be adapted to other products. The project concludes with a documented lab-scale integrated succinic acid–ISPR system, demonstrated performance improvements versus the reference process, and a first set of transferable design rules.

4. R&R (Roles and Responsibilities)

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Expected R&R of Research Institute	**Technical contributions & research scope** Fraunhofer UMSICHT will be responsible for the development, operation and optimization of the succinic acid fermentation at lab scale, primarily in 1–5 L stirred-tank bioreactors, with the option to use larger volumes where appropriate. The institute will integrate the Korean ISPR modules into a side-stream fermentation loop, including the hydraulic layout, necessary instrumentation and basic process control. It will systematically investigate the impact of ISPR on process performance (titer, productivity, yield), base consumption, salt formation and culture stability. Furthermore, Fraunhofer UMSICHT will develop and test simple, biology-based (semi-)automated control strategies for product removal (e.g. pH- or conductivity-triggered operation) and contribute to process modelling and platform development by establishing simple dynamic models and design rules for transferring the concept to other organic acid fermentations. **Specific tasks** Within this scope, Fraunhofer UMSICHT will establish and document a reference succinic acid fermentation as a baseline for comparison. It will provide real fermentation broths to the Korean partner for separation tests and specifications. The institute will design and build the integrated lab setup combining bioreactor and ISPR side-stream, including pumps, valves, sensors and control hardware/software. It will plan and execute integration trials as well as multi-day runs of the combined system and generate and analyze experimental data required for model calibration and for the derivation of practical operating guidelines. **Resources to be provided by Fraunhofer UMSICHT** Fraunhofer UMSICHT will provide access to bioprocess laboratories and pilot infrastructure, including multiple bioreactor scales and standard analytical capabilities (e.g. HPLC/IC). Dedicated scientific and technical staff will be assigned for fermentation, process integration, automation and data analysis. In addition, the institute will make available its existing process control and data acquisition systems as well as computational tools for basic process modelling and evaluation.
Expected R&R of Korean SME Partner	**Technical contributions and development tasks** The Korean SME will lead the design, development and optimization of in-situ product removal (ISPR) concepts for organic acids, using succinic acid as the primary use case. This includes the selection and/or development of suitable separation materials and modules (for example membranes, adsorbents/ion exchangers or extraction systems) that are tailored for selective organic acid removal with minimal co-removal of sugars and essential nutrients. The SME will engineer lab-scale ISPR modules (such as cells, cartridges, columns or contactors) that are compatible with bioprocess requirements, in particular with regard to sterility, cleanability and acceptable pressure drops. **Testing and validation activities**

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	The SME will perform laboratory screening of separation principles using synthetic solutions (organic acid plus salts, later including simple sugar mixtures) to quantify separation performance in terms of capacity, selectivity, flux or cycle times, and to determine sensitivity to pH and ionic strength. This will be followed by extended testing with real fermentation broths supplied by Fraunhofer UMSICHT, including the assessment of fouling behavior, stability, cleaning strategies and long-term operation. Based on these studies, the SME will deliver characterized ISPR modules for integration into the Fraunhofer bioreactor setup, together with operating guidelines and recommended parameter ranges. **Resource commitments** To fulfil these tasks, the Korean SME will provide qualified R&D staff, including process engineers and separation specialists, to design, build and test the ISPR modules. It will make available its laboratory and pilot-scale separation infrastructure, such as test rigs, pumps, control systems and analytical equipment, for concept evaluation and module optimization. The SME will also supply prototype ISPR modules and key spare parts to Fraunhofer UMSICHT to enable integration and long-term testing of the modules within the bioprocess environment. **Collaboration and support efforts** The SME will maintain close technical coordination with Fraunhofer UMSICHT regarding module specifications, interface design and operating conditions through regular joint meetings, data exchange and joint interpretation of results. It will provide support during integration and troubleshooting of the modules in the fermentation setup, including remote assistance and, where necessary, on-site visits to adjust module operation and resolve technical issues. In addition, the SME will jointly contribute to documentation, publications and project reporting, particularly on the performance and applicability of the ISPR modules and their potential transfer to other organic acid processes.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint R&D project is expected to deliver a validated, lab-scale integrated ISPR–fermentation system for succinic acid (reactor volume: 1–5 L, with optional confirmation runs up to 30 L at Fraunhofer UMSICHT) that clearly outperforms the current reference process in terms of achievable titers, productivities, and reduced base and salt consumption. One or two well-characterized ISPR module types for organic acids will be developed, with demonstrated selectivity for succinic acid, limited co-removal of substrates and nutrients, and proven stability and cleanability in contact with real fermentation broths. In addition, the partners will establish simple, biology-based (semi-)automated operation of the integrated system (e.g. pH-triggered product removal) and derive basic dynamic models and design guidelines that support optimization and transfer to other organic acid processes. For the Korean SME, the results provide a concrete basis for commercial ISPR module products for in-situ product removal, supported by performance data and a clear use-case. For Fraunhofer UMSICHT and European stakeholders, the project delivers a practical blueprint for intensified organic acid fermentations that can be used in follow-up industrial development and extended beyond succinic acid. Overall, the outcomes have the potential to lower downstream processing costs, improve reactor and utility efficiency, and strengthen Korea–Europe collaboration in separation and bioprocess technologies, thereby contributing to more competitive and sustainable bio-based chemical value chains.

6. References

(1) BMBF-funded project (2024-2027): “Biolectid” – Innovative Approach to utilize CO_2 in a new bioelectrochemical fermentation process for succinic acid production

(2) Dissertation (2018): Geppert, Florian – Development of a bioelectrochemical fermentation reactor system

(3) Research Group (2006–2011): “C4-Gain”-Research group for biobased, fermentative production of succinic acid

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization